

Submission DSA1-FINAL-SUB04

Question DSA1-FINAL-Q01

1st step / setup:

$7 > 4$, go right; $7 < 9$, go left; $7 > 6$, go right.

then substitute ...

= _____ (ran out / not finished)

Question DSA1-FINAL-Q02

1st step / setup:

Maintain a hash set of values seen so far.

then substitute ...

= _____ (ran out / not finished)

Question DSA1-FINAL-Q03

1st step / setup:

A balanced tree can be built recursively by choosing the median, with each key visited once: $\Theta(n)$.

then substitute ...

= _____ (ran out / not finished)

Question DSA1-FINAL-Q04

1st step / setup:

Queue A; discover B,C; then dequeue B and discover D; dequeue C and discover E.

then substitute ...

= _____ (ran out / not finished)

Question DSA1-FINAL-Q05

1st step / setup:

Invariant: the prefix before index i is sorted and contains the original prefix values.

then substitute ...

= _____ (ran out / not finished)

Question DSA1-FINAL-Q06

1st step / setup:

$7 > 4$, go right; $7 < 9$, go left; $7 > 6$, go right.

then substitute ...

= _____ (ran out / not finished)

Question DSA1-FINAL-Q07

1st step / setup:

Maintain a hash set of values seen so far.

then substitute ...

= _____ (ran out / not finished)

Question DSA1-FINAL-Q08

1st step / setup:

A balanced tree can be built recursively by choosing the median, with each key visited once: $\Theta(n)$.

then substitute ...

= _____ (ran out / not finished)

Question DSA1-FINAL-Q09

1st step / setup:

Queue A; discover B,C; then dequeue B and discover D; dequeue C and discover E.

then substitute ...

= _____ (ran out / not finished)

Question DSA1-FINAL-Q10

1st step / setup:

Invariant: the prefix before index i is sorted and contains the original prefix values.

then substitute ...

= _____ (ran out / not finished)